pop^n $\sim F(\mu, \sigma^2, \ldots)$ →

goal: Infer (estimate) $\mu, \sigma^2, \gamma \ldots$ based on Samples

$x_1 \ldots x_n$ point

Problem → you can get estimates of μ. ✓

but se, $\hat{CI}$, p-values, bias → X

↑ ↑ ↑ ↑

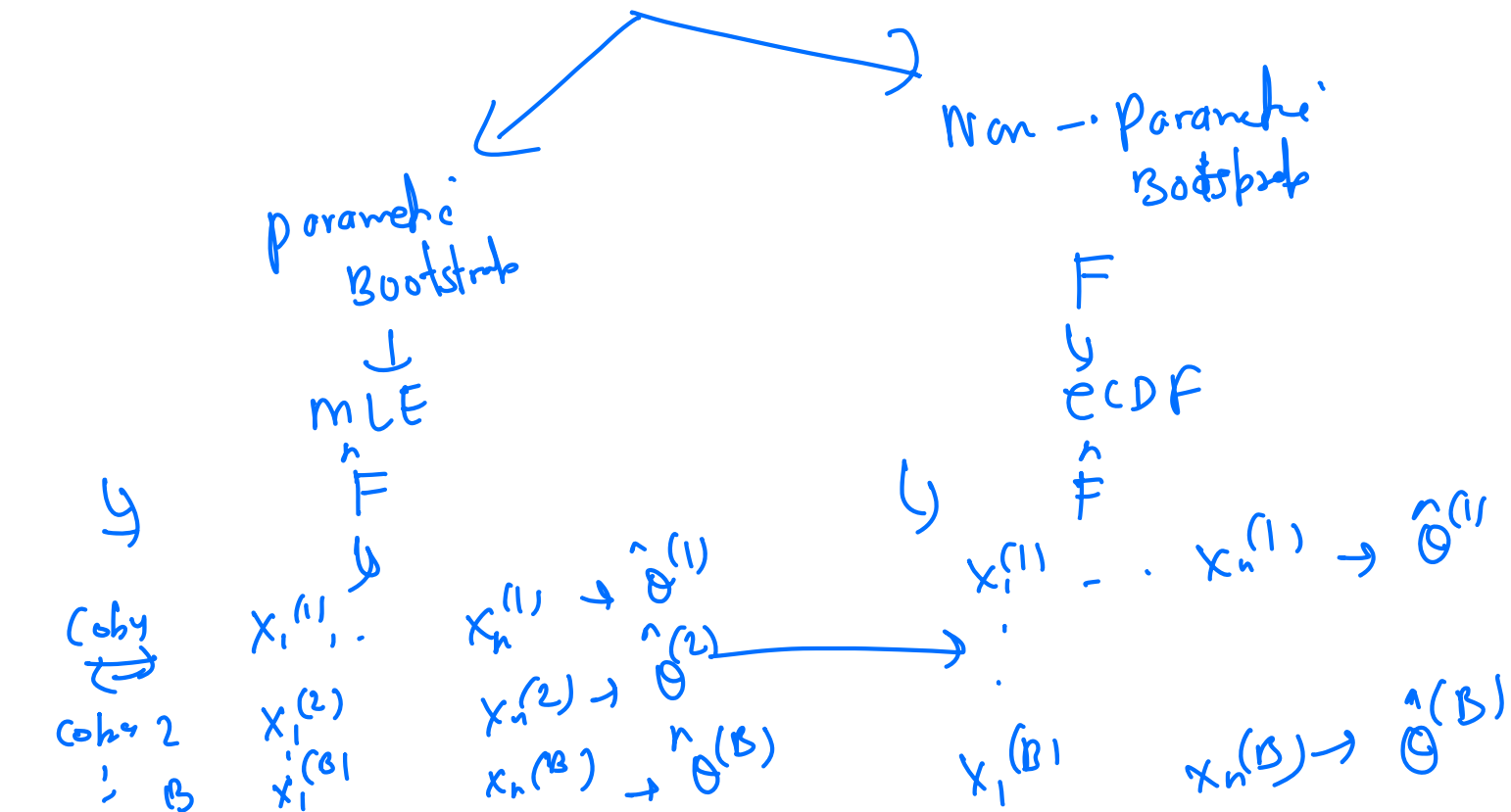
an possibility ↳ multiple copy of $x_1 \ldots x_n$

→ $\bar{x}_1$, $\bar{x}_2$, ⋮, $\bar{x}_{10}$ → se $(\bar{x})$, $CI(\bar{x})$ ↳

↳ Bootstrap → $x_1 \ldots x_n$

create multiple copy of existing sample $x_1 \ldots x_n$

↳ 2 possible



The Jackknife: Leave One Out

- Instead of random resamples, use the n **deterministic** leave-one-out samples

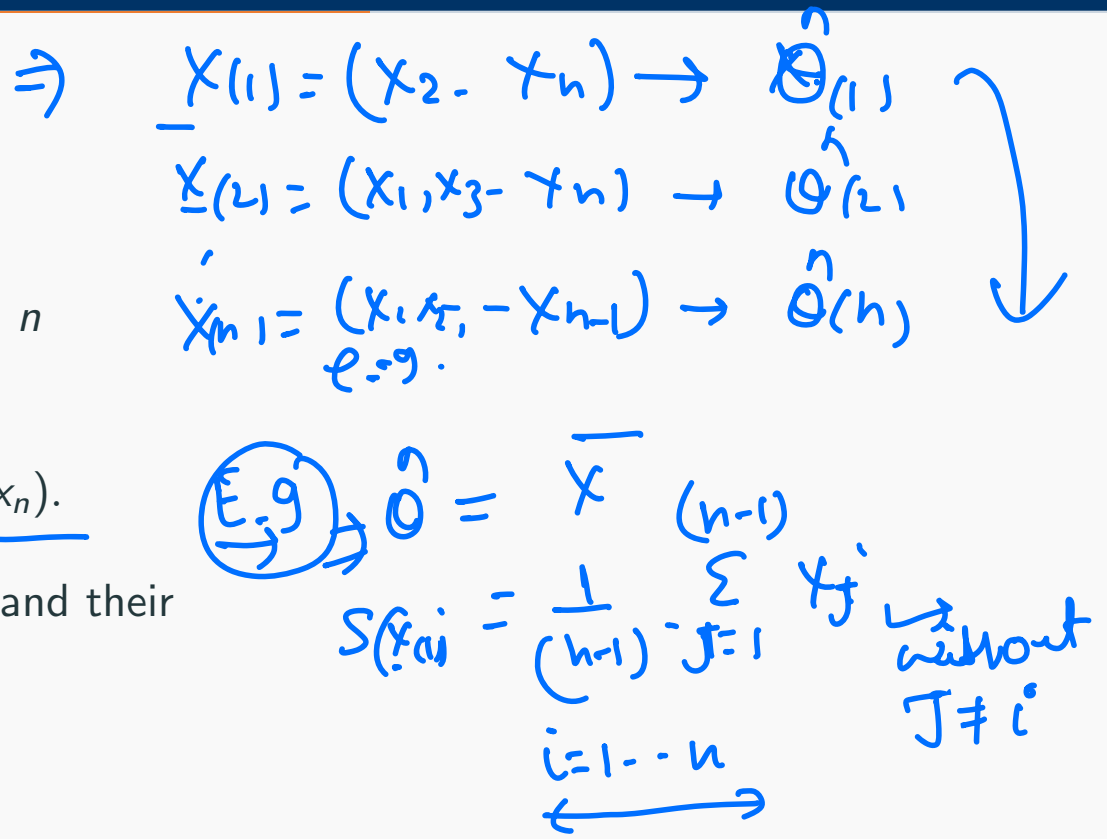
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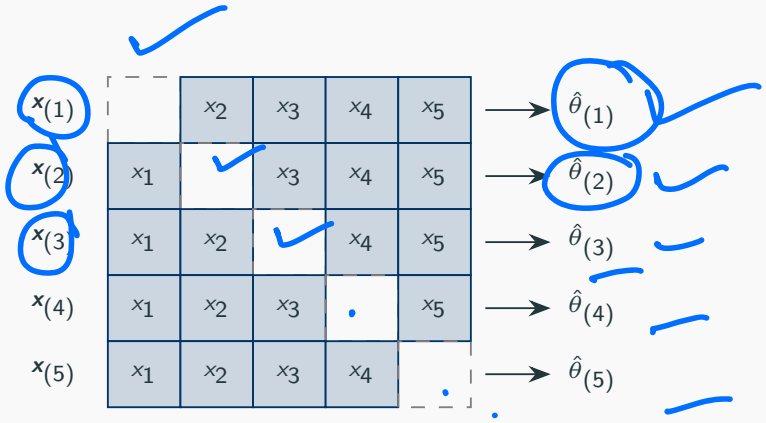


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Jackknife estimates

$$\widehat{\text{bias}}_{\text{jack}} = (n-1)\big(\hat{\theta}_{(\cdot)} - \hat{\theta}\big), \qquad \widehat{\text{se}}_{\text{jack}} = \sqrt{\frac{n-1}{n}\sum_{i=1}^{n}\big(\hat{\theta}_{(i)} - \hat{\theta}_{(\cdot)}\big)^2},$$

and the bias-corrected estimator $\hat{\theta}_{\text{jack}} = n\hat{\theta} - (n-1)\hat{\theta}_{(\cdot)}$.

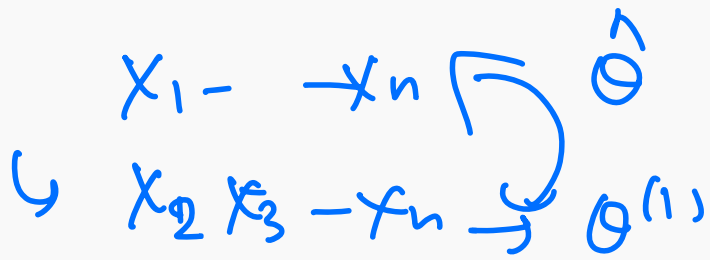
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Two checks you can do by hand

- **Sample mean:** $\hat{\theta}_{(i)} = (n\bar{x} - x_i)/(n-1)$, which gives $\widehat{\text{se}}_{\text{jack}} = s/\sqrt{n}$ *exactly*.
- **Plug-in variance** $\hat{\sigma}^2$ (divisor n): the jackknife bias correction returns s^2 (divisor $n-1$) *exactly* – it removes the $O(1/n)$ bias.

Check 1 (Mean): why $\widehat{\mathrm{se}}_{\mathrm{jack}} = s/\sqrt{n}$

Let $\hat{\theta} = \bar{x}$ and $\hat{\theta}_{(i)} = \bar{x}_{(i)}$.

removing ith sample

$$\bar{x}_{(i)} = \frac{1}{n-1}\sum_{j\neq i} x_j = \frac{n\bar{x} - x_i}{n-1}.$$

e.g.: ⇒ interest → μ: population mean

↳ estimator: $\bar{X} = \hat{\theta}$

Bias se

$$\bar{X}_{(i)} = \frac{1}{(n-1)}(X_1 + X_2 + \cdots X_{i-1} + X_{i+1} \cdots + X_n)$$

$$(n-1)\bar{X}_{(i)} = X_1 + \cdots + X_{i-1} + X_{i+1} + \cdots + X_n + X_i - X_i$$

$$= n\bar{X} - X_i$$

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$$= \frac{1}{n-1}\sum_{i=1}^{n}\left(\frac{n\bar{x}}{n} - \frac{x_i}{n}\right) = \frac{1}{n-1}\left(n\bar{x} - \frac{1}{n}\sum_{i=1}^{n} x_i\right) \to \bar{x}$$

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$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$

Plugging into the jackknife SE formula gives

$$\widehat{\mathrm{se}}^2_{\mathrm{jack}} = \frac{n-1}{n}\cdot\frac{1}{(n-1)^2}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{1}{n(n-1)}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{s^2}{n}.$$

$\widehat{\mathrm{se}}^2(\mathrm{jack}) = \frac{n-1}{n}\sum_{i=1}^{n}(\hat{\theta}_{(i)} - \hat{\theta}_{(\cdot)})^2$

Where the Jackknife Fails – and Bootstrap vs Jackknife

Failure for non-smooth statistics

For the **sample median**, deleting one point moves $\hat{\theta}_{(i)}$ to one of only two values, so $\widehat{\text{se}}_{\text{jack}}$ is *inconsistent* – it does not converge to the true standard error. (The delete-d jackknife with $d \to \infty$ repairs this.)